LS-CAT Management Board Meeting

Connection Instructions

To join the meeting on a computer or mobile phone:

https://northwestern.zoom.us/j/91778215682

Or for just the audio:

(312) 626-6799

Meeting ID: 917 7821 5682

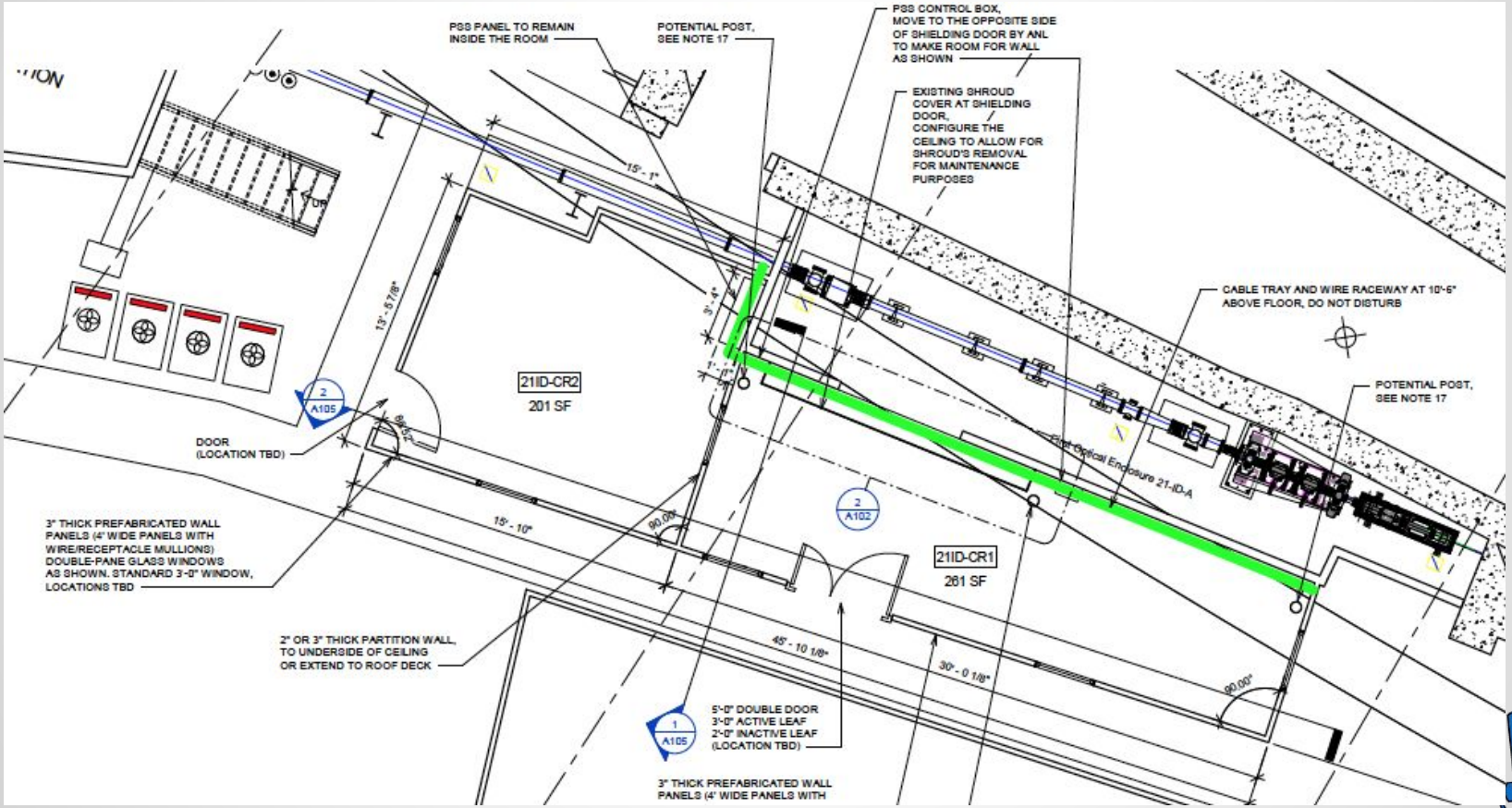
Agenda

9:45	Sound Check / Informal Greetings
10:00	Welcome: Minutes and Agenda Approval (Stuckey)
10:10	LS-CAT Status Update (Brister) ● Operations in the COVID-19 era ● Monochromator maintenance ● New Offices ● Upgrade Planning Update ● S10 Proposals Update (Alfonso) ● New LS-CAT Agreement Status
10:35	LS-CAT FY 2021 Budget
10:45	New Business
10:50	Adjourn

LS-CAT Status

- APS starting to allow non-COVID-19 experiments.
  - Remote access only
  - Still in minimum safe operations mode: LS-CAT is operating as we do on weekends: only one staff member on-site except for equipment repairs or preparations for the new control rooms when a second person is allowed
  - Expect to stay in this mode for the duration of the next run
  - The APS moved the May and September Down periods around so that at least the APS or NSLS II would be up and running. The May shutdown now lasts only 10 days and the September shutdown starts in mid August and is included in the 2020-2 run cycle. There are only a couple of days between 2020-2 and 2020-3 cycles. Expect a bit of confusion.
- Monochromator work planned for the May shutdown moved to September
  - 21-ID-D may have restricted energy ranges until repairs are completed
  - 21-ID-F/G work must be completed before coming up from the dark period

New LS-CAT Office Space (Coming this summer)



APS Upgrade

- Work on APS-U critical path items has continued during the COVID-19 Minimum Safe Operations Mode
- Relocating LS-CAT offices are on the critical path
- Official start of dark period: June 2022
- Official commitment to June 2022 date: January 2022
- LS-CAT commitment to FY 2022 Budget: May 2021
- There are undercurrents of a delay that pushes start date to January 2023: we do not expect the official party line to change until January 2022

LS-CAT Upgrade Planning List (Partial)

		Priority	Funded	Worst Case	Item	Est Cost	Contingency	Worst Case Spending
Total of all Costs	$9,358,000	1	1	1	Upgrade main storage to 1PB	40,000	10,000	50,000
Maximum Priority	10	2	1	1	Move Inboard Beam masks downstream and lengthen bellows	10,000	2,500	62,500
Priority Costs	$928,000	3	1	1	Remove side-bounce mono's water to vacuum braze (2 each)	20,000	1,000	83,500
		4	1	1	Service Cryocooler (1 each)	5,000	1,250	89,750
Worst Case Cost Contingency	25%	5	1	1	Service Cryo-Tigers (10 each)	75,000	18,750	183,500
		6	1	1	Service Cryo-Jets	5,000	1,250	189,750
Funding Level	$1,000,000	7	1	1	Upgrade Motor Controllers	50,000	12,500	252,250
		8	1	1	Service D-Mono Electro-Mechanics	33,000	8,250	293,500
		9	1	1	Replace D diffractometer with MD3-UP	650,000	20,000	963,500
		10	1	0	New D-Mono Cooling and Crystals	40,000	10,000	1,013,500
		11	0	0	Replace F Diffractometer with MD2S	460,000	20,000	1,493,500
		12	0	0	Replace G Diffractometer with MD2S	460,000	20,000	1,973,500
		13	0	0	New Cryocooler	300,000	75,000	2,348,500
		14	0	0	New Cryojets	300,000	75,000	2,723,500
		15	0	0	New K-B Mirrors and benders for 21-ID-D	300,000	75,000	3,098,500
		16	0	0	Purchase 2 new detectors	4,200,000	200,000	7,498,500

S10 Proposals Status



MD3-UP Micro Diffractometer
Requested for 21-ID-D



Eiger-2 X 16M Requested for 21-ID-D
(Current Eiger detector would move to 21-ID-F)

Progress Toward a new LS-CAT Agreement

Forest and Brister worked through the full LS-CAT agreement making modifications. Some questions for the NU lawyers remain. We are still on track to having a new agreement ready by October 2020.

Remaining Details

- Can we have each institution sign in parallel instead of in series?
- How to keep from changing the agreement as the membership changes?
- Liability coordination

Modified LS-CAT Agreement

FY 2020 Budget and Predicted FY 2021 Budget

FY20 Budget: Current budget as approved last spring

FY20 Total (Est): Best guess of how things are going based on the first 8 months of the fiscal year (which started September 1)

FY21 Proposed: The proposed budget for the upcoming fiscal year.

Description	FY2020 Budget	FY2020 Total (Est)	FY 2021
Salaries	871,201	858,905	867,428
Fringe Benefits	245,679	245,583	259,361
Supplies	40,000	34,026	40,000
Supplies, Restricted	2,500	1,008	2,500
Services	30,000	16,115	20,000
Services, Restricted	20,000	18,633	20,000
External Space Rental	135,242	123,535	0
Travel, Domestic	5,000	544	0
Utilities & Telephone	9,000	6,211	9,000
Capital Equipment	34,140	142,115	86,588
Indirect Cost - Sponsored	398,316	296,922	396,647
Total	1,791,078	1,743,597	1,701,524
Per Share	358,216		358,216

LS-CAT Budget for FY 2021

Salaries				100%	KB, MB, JVO, SA, JSB, EK, ZW
	Base	867,428		50%	MG
	Fringe	259,361			
	Total		1,126,789		
Items F&A Exempt					
	Capital Equip.	86,588			
	Rent	0			
	Telcom	9,000			
	Total		95,588		
Non-Salary Items Subject to F&A					
	Lab Supplies	40,000			
	Office Supplies	2,500			
	Lab Services	20,000			
	Insurance	20,000			
	Travel	0			
	Total		82,500		
F&A			396,647		
Total			1,701,524		
Per Share			340,305	NU UM MSU	
Per 1/2 Share			170,152	UW UI VU	
Per 1/4 Share			85,076	WSU	

Vote 1:
Approval of amounts shown.

Vote 2:
Exact amount of capital equipment needed depends on what breaks and which upgrades cannot be postponed. Monies left on this line will be added to capital fund or, conversely, when outlays exceed this amount we'll dip into the capital fund. As always, the board will be kept informed about unusually large expenses. Vote 2 allows the capital equipment funds to be used for normal upkeep of the facility.

New Business